Scalability & Future Plan

Date	30 September 2026
Team id	SWTID-2026-2551
Project name	Fit Buddy – AI Fitness Plan Generator using Gemini Models
Maximum mark	1 marks

Scalability & Future Plan

This document outlines how the **Fit Buddy – AI Fitness Plan Generator using Gemini Models** solution can be scaled to support a larger number of users, increased fitness-related data, and additional features in the future. It also captures the roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Dependence on Gemini API availability and response time	AI fitness plan generation may be affected by API availability or network conditions.	High
2	Limited scalability for a large number of concurrent users	Increased users may result in higher server load and longer response times.	High
3	Limited fitness data and personalization capabilities	More user-specific data and advanced personalization may be required for future improvements.	Medium

Scalability Plan

s.no	Scalability aspect	Current state	Proposed upgrade /solution
1	User load	Suitable for a limited number of users	Use cloud deployment and scalable server resources

2	Data storage	Basic storage for user fitness information	Use a scalable cloud database
3	Performance	Suitable for normal usage	Implement caching, optimized APIs, and faster processing
4	Security	Basics user-data protection	Add authentication, encryption ,and secure API-key management

Future Roadmap

Planned Feature / Enhancement	**Target Timeline**	**Expected Impact**
Improve personalization of AI-generated fitness plans	Short Term	Provide fitness plans more closely aligned with individual user requirements.
Fitness progress tracking and plan history	Medium Term	Allow users to monitor their fitness progress and review previous plans.
Mobile application and wearable-device integration	Long Term	Enable users to access Fit Buddy on mobile devices and use activity data for better personalization.
Advanced AI fitness recommendations and analytics	Long Term	Provide more detailed fitness insights, adaptive recommendations, and improved user experience.

The uploaded template specifically covers **Current System Limitations**, **Scalability Plan** across user load/data storage/performance/security, and a phased **Future Roadmap**.

Sure. Below is the **same Team Involvement in Demonstration template** adapted for **Fit Buddy – AI Fitness Plan Generator using Gemini Models**, while preserving the original structure and fields.